



# Fishing as a livelihood, a way of life, or just a job: considering the complexity of “fishing communities” in research and policy

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**Abstract** In the scientific literature on fisheries, the concept of *community* is often used broadly to indicate a place-based group whose members are dedicated to fisheries and have relatively homogeneous economic, social, and cultural interests. However, this categorical perspective to scope a “fishing community” is not necessarily an insightful approach to explore diverse social relationships with the marine environment, fishwork, and management in a practical context, and risks mismatches with science-based recommendations for management and policy. Drawing from ethnographic work, we highlight different historical and cultural dynamics from four case studies from fisheries in northwest Mexico. We identify key factors that help contextualize fishwork relationships, related to the importance of fishing practices on worldviews, daily routines, and income. These are used to derive three configurations (livelihood, way of life, and job) that describe and give analytical content

to the notion of these fishing communities. Our use of a typology is not intended to generalize them or provide universal categories, but rather to convey to a broad range of fisheries scientists the importance of considering social contexts in the places in which we work and learn, and a set of guiding questions that may help in this regard. Contextualizing the importance of historical and cultural factors in scoping community units beyond occupational or geographical characteristics is essential for identifying and addressing (in)equitable processes and outcomes in fisheries sectors, research, and management.

**Keywords** Artisanal fisheries · Local communities · Indigenous communities · Livelihoods · Social-ecological systems · Mexico

## Introduction

The scope of mainstream fisheries research has broadened from stock-based assessments and management, to ecosystem approaches, to the socio-economic dynamics embedded in fisheries practices and ecosystem changes. This increased focus on the human dimensions of fisheries has led to a common framing of fisheries research in terms of effects on “fishing communities.” However, to researchers specifically focused on artisanal or small-scale fishing practices, the versatility with which the notion of ‘fishing community’ is used is often confusing

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and sometimes problematic because of its undefined nature and fluid terminology (Brookfield et al., 2005). It has indeed been argued that this lack of focus has led to misunderstandings on social context and subsequent inappropriate or otherwise ineffectual policies and management strategies to address what are often complicated social-ecological issues (Syed et al., 2018; Jentoft and Chuenpagdee, 2009).

This paper draws from ethnographic research in Northern Mexico to provide an example of how researchers can consider relationships within “fishing communities” in ways that accommodate diverse social and economic dynamics; we argue this can help research better inform policy design and management implementation strategies. By unpacking the dynamics embedded in an ethnographic context, we provide an example typology of ‘fishing community’ that a broad range of fisheries scientists could utilize to deepen their understanding of the importance of historical and cultural contexts in research and management recommendations. Our use of a typology is certainly not intended to generalize all communities or provide universal categories, but rather to aid a broad range of fisheries scientists in considering diverse social contexts in the places in which we work and learn.

There is a large and rapidly increasing focus specifically on management and policy within the fisheries research field (Syed et al., 2018), which is perhaps positive for providing scientific evidence to inform policy but risks leading to inequitable impacts on fishers when recommendations lack nuance or social context (Syed et al., 2018; Song et al., 2020). This can be a particular challenge given the inherent diversity of fisheries around the world, and particularly artisanal and small-scale fisheries that tend to be more place-based (Jentoft et al., 2017). For example, the use of ‘fisheries-dependent communities’ in European Union (EU) aid allocation—and its definition based on quantitative metrics such as a given percentage of fishworkers—can be highly problematic when individuals are increasingly diversifying income sources but remain culturally attached to the practice of fishing (Brookfield et al., 2005). In the US, similar arguments have been made regarding the mismatch and repercussions of official definitions of “fishing community”—again based on economic participation metrics—compared to much more nuanced and fluid concepts based on anthropological research that are

rarely documented in wider research or considered in practice (Clay and Olson, 2008).

In less developed regions, the shorthand aim of ‘benefits to fishing communities’ (e.g., Cline et al., 2017) fits with what are commonly more collectivist approaches to fisheries management, but reviews show that published research rarely delves into social-ecological relationships (Syed et al., 2018) or defines who exactly these communities represent (Delgado et al. *in press*). This exclusion by generalization ignores specific populations and power imbalances, and risks further marginalizing individuals with little input into fisheries policy, including women, Indigenous fishers, and fishers without permits (who may or may not be locally considered as ‘illegal’) (Finkbeiner et al. 2017).

An overall shift in research approach to this critical issue is clearly required given the risk of impacts as outlined above. This is also important given the recognition of the fisheries sector—and social dynamics more broadly—as central to global efforts to establish equitable and sustainable Blue Economies (Cisneros-Montemayor et al., 2021) building from directed scientific research during the ongoing UN Decade of Ocean Science (Singh et al. 2021). Fishing is traditionally the occupational core of many marine coastal communities and the relevance of social and cultural dimensions of fishing activities is clear: *“the fishing industry is seen to be the forum through which communities’ bonds, values, knowledge, language and traditions are established, confirmed and passed on. [...] is the ‘way of life’ for the community [...] fishing is the glue that holds the community together”* (Brookfield, et al., 2004: 56). Despite the issues discussed above regarding the common unspecific use of ‘community’ in fisheries management literature specifically, the broader concept of community has been extensively discussed and defined and indeed can be formally framed under key characteristics including material conditions, sense of totality, and homogeneity in the sense of exclusive cultural consciousness (Trápaga; 2018). Community also commonly implies basic elements such as belonging, being together, and social interaction (Tjora y Scambler, 2020). These contexts must be integrated into our understanding and use of the term to clearly and transparently define units of research and to relate social-ecological dynamics that should be reflected in fisheries policy and management strategies.

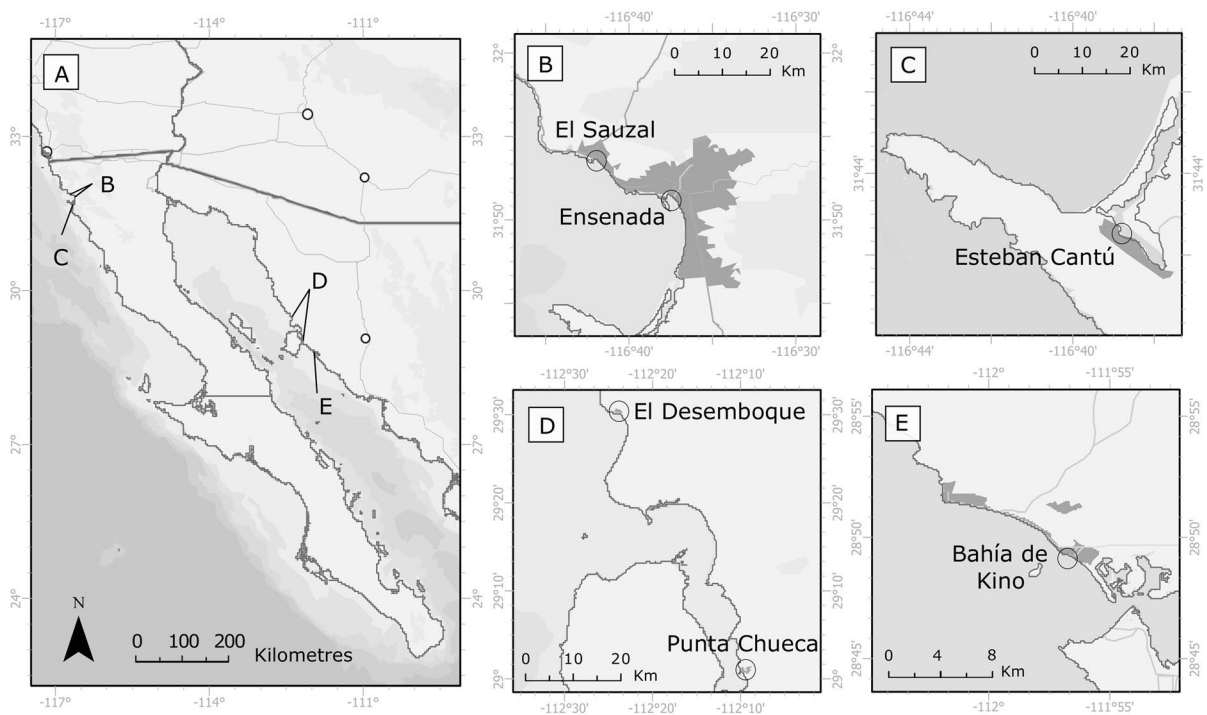
Four case studies highlight the complexity and heterogeneity of people dedicated to fishing activities in the Mexican northwest and a formal conceptualization of them can aid researchers throughout the world. Following from the above, this paper provides a framework for conceptualizing and discussing various types of relationships between humans and fishing, using four case studies in Mexico to ground and illustrate the analysis. The objective of this analysis and our proposed framework is to help researchers in both social and natural sciences to bound and better define ‘fishing communities’ where they work, particularly with a view to informing recommendations for fisheries and environmental policy.

## Methods

The empirical basis for the analysis of the case studies in this paper is ethnographic fieldwork including direct and participant observation, formal and informal interviews, and participatory workshops. This work was carried out in situ from 2018 to 2021 in

the communities described below (see Fig. 1), and is explained at length in Delgado (2019, 2021). Our methodology followed a multiple embedded case study approach, where physical localities can include more than one sub-unit of analysis (Yin, 1994), and included three stages.

The first stage was exploratory and included a review of publications expressly focused on ‘fishing communities’ in Latin America and the Caribbean (140 peer-reviewed, 25 grey literature; Delgado et al., in press). The second stage was descriptive and developed the case studies, exploring through ethnography how individuals establish relationships with fisheries-related work and how these relationships help configure communities at different levels. Fieldwork involved ethnographic methods to observe and compile first-hand qualitative information following a ‘thick description’ approach that seeks to document social activities and their broader socioeconomic and ecological contexts in ways that can be understood by outsiders to these social groups (Geertz, 2004; Marsvati, 2004). This most often involves longitudinal research, and extensive (2–4 month) field research



**Fig. 1** A Study area in northwest Mexico and case studies of ‘fishing community’ including B El Sauzal-Ensenada, C Esteban Cantú, D El Desemboque-Punta Chueca, and E Bahía de Kino

was carried out in Sonora sites yearly from 1998 to 2006, with intensive (2–4 week) field research in 2009 and 2018. In Baja California sites, extensive research was carried out yearly from 2012 to 2016, and intensive yearly from 2018 to 2021. This field work involved:

- (a) Direct observation of daily relations between fishwork sectors in the case study localities;
- (b) Open interviews with fishers and divers, members of their families, and individuals in the localities, both associated and not with fishwork;
- (c) Group participatory workshops in Baja California sites.

The third stage was analytical and used case ethnographies together with other bibliographical and sources to triangulate and update relevant information, validate both published and empirical information, and contextualize the history and context of these villages, as described in depth below. Given the embedded nature of ethnographic fieldwork, we also include direct quotes in the results which complement and enrich the historical and quantitative narratives, and lead to the formulation of our proposed examples of types of relationships with fisheries. While we argue that such in-depth research is highly useful and likely essential for fully capturing the complexity of communities (including those where fishing is a primary activity), we mainly aim to highlight these complexities to support our advice regarding useful frameworks and a need for baseline efforts to better define ‘fishing community’ in fisheries research and its recommendations for policy.

## Results

### Regional context of the case studies

Industrial fishing in Mexico is concentrated in a few major ports distributed in various states, while artisanal or *ribereña* fisheries are ubiquitous in any coastal rural localities and cities and often coexists with industrial fishing in larger ports. (*Pesca ribereña* is the most common way to refer to artisanal and small-scale fishing activity in Mexico. It is not specific to a gear or target species, but generally takes place aboard *pangas*, ~7 m open-deck fiberglass vessels

with outboard engines and usually 2–4 fishers.) Artisanal fishing predominates, occupying 85% of fishers in the country and more than 90% of total boats (Jiménez et al. 2018). In addition, although industrial fisheries receive more political attention and financial support, artisanal fisheries produce the majority of seafood for national consumption and have a higher total economic value (Schuhbauer et al., 2020). A few rural localities are composed mostly by Indigenous populations with a historical and cultural cosmogony entwined with the sea and its resources. Here, fishing tends to be oriented to household consumption, with a smaller portion exchanged or sold at local markets (Espinoza et al., 2011). As we highlight in these case studies, women are key actors in fisheries throughout Mexico (as elsewhere in the world), including as fishers and owners of fishing licenses, and most commonly in the processing and sale of catches (Blázquez and Palacios, 2016).

The Mexican northwest includes the states of Sonora, Sinaloa and Nayarit, as well as the states of Baja California Sur and Baja California on the Baja California Peninsula (Fig. 1). While fisheries take place across the entire country, this region contributes 70% of national annual production and tends to dominate national marine research and conservation efforts (Palacios-Abrantes et al., 2019). Figure 1 shows our case study localities, Ejido Esteban Cantú, Ensenada-El Sauzal, Bahía de Kino, and El Desemboque-Punta Chueca.

### Ejido Esteban Cantú, Ensenada, Baja California

The *ejido* (in México, communal land tenures owned by a corresponding group of *ejidatarios*) Coronel Esteban Cantú (pop. 545; INEGI, 2020) is located outside the city of Ensenada in a small peninsula called Punta Banda (Fig. 1 C). Its population organized at the end of the 1960s to request property rights for the land they already occupied and by 1971 were granted over 10 thousand hectares by the federal government, distributed among 89 *ejidatarios*. A group of 23 men among these *ejidatarios* had already fished there for at least three decades, targeting lobster, abalone, and finfish. In 1994, they organized themselves as a Society of Rural Production Divers and Fishers from the *Ejido* Coronel Esteban Cantú (henceforth, SPR) with 39 members, 23 *ejidatarios* and 16 fishers from neighboring localities. At least half of the

members are linked by kinship or marriage (Delgado, 2019).

These divers and fishers now mainly target red and purple urchin (*Mesocentrotus franciscanus*, *Strongylocentrotus purpuratus*), sea cucumber (*Parastichopus parvimensis*), starfish (*Pisaster* spp.), and fin-fishes. Sea urchin is the most important fishery and is entirely exported to Asian markets, while other fisheries provide additional seasonal income throughout the year. Processing of catch (breaking, cleaning and packaging the urchin gonads) within the SPR offers around 20 jobs to women and men in Punta Banda. The income that fishers and divers receive from the urchin fishery is relatively high and allows them to fully cover domestic expenses while their spouses are mainly dedicated to household work and childcare (Delgado, 2021).

Women primarily work in processing and sales, though some may work occasionally in capture fisheries, particularly for starfish. Nevertheless, when members of the SPR pass away, their “membership” is transferred to their wife, who generally hands it to a son to work in his father’s place. However, when there are no male persons that can or want to work it, widows can lease the place out to a man recommended by other members of the organization, generally the son another member. Income to the fisher is divided into his part and the widow’s part. Through these agreements, members of the SPR organization continue to primarily be people from within the *ejido*, even when some active workers are external.

The SPR has two management polygons for urchin capture of urchin, cucumber, and starfish. Members also have multi-use fragments of *ejido* land which is used in three main ways: 1) divers and active fishers dedicating their free time to low-impact tourism and/or beekeeping; 2) divers and fishers that are only active during urchin season (July to February) and that dedicate the rest of the year to agriculture and/or small-scale livestock care; and 3) inactive divers and fishers that lease their places in the SPR and are fully dedicated to agricultural and livestock work or another activity. During seasonal closures for red sea urchin, it is common that three or four divers from the SPR are hired as industrial maintenance divers in tuna mariculture ranches in nearby Ensenada (Fig. 1).

Although the SPR does not provide health care or a retirement pension system for its members, it can provide emergency funds, for example covering the

funerary expenses of one of their members or workers. Sometimes, the SPR sponsors uniforms for Esteban Cantú’s amateur baseball team or celebrations organized by the *ejido*, to which they usually invite *ejidatarios* and *avecindados* (neighbors) that reside in the town. *Ejidatarios* and neighbors have a daily relationship with the city of Ensenada (Fig. 1), where shops, banks and government offices are located. Members of the directive council of the SPR also go to make any transaction relative to the fishing activity, to buy supplies and to attend meetings with other fishing organizations, fisheries researchers and governmental agents that integrate the Urchin Product System in Baja California (government management units regulating key fisheries, in this case sea urchin). Nevertheless, their immediate daily life and personal relationships take place within Esteban Cantú).

#### Port of Ensenada: El Sauzal de Rodríguez, and Ensenada

During the 1930s, Abelardo Rodríguez bought and installed abalone and lobster canneries in the north of the city of Ensenada (Fig. 1 B) in what is now El Sauzal de Rodríguez township. Abalone and lobster production had increased since the end of XIX century, supported by American investment and the training of divers and fishers by Japanese men that were developing commercial abalone, lobster, sardine and tuna fishing between the years 1912 and 1941 (Velázquez, 2007).

Rodríguez turned this ‘Marine Products Company’ into a parastatal company named *Pesquera del Pacífico* and also developed a fishing “company town” (Zapata, 1977) 4 miles from Ensenada harbor. This company town included a health clinic, school, baseball team, and homes for fishers, divers and their families. This habitational zone became what is now the Manchuria neighborhood. The men worked in capture, while the women were employed in the packing and canning factories. In 1939, once the Ensenada-Tijuana highway was completed, connecting the region with California, the neighborhood became a growing hub of commercial activity. In addition to fishworkers, El Sauzal became home for longshoremen and shipbuilders working at the shipyard, thus comprising individuals from throughout the seafood value chain.

Most of the Baja California industrial fleet is still located in the harbor at El Sauzal and the Manchuria neighborhood still exists, surrounded by seafood packing factories, aquaculture enterprises and manufacturing plants. New residential neighborhoods are being developed for upper-middle income families, from which fishers and their families are totally excluded. However, many residents are still industrial fishers dedicated mainly to the sardine fishery (*Sardinops sagax*), as well as small-scale fishers and fishworkers engaged in fisheries for lobster, finfishes and sea urchin, and the packaging of marine products from both fisheries and aquaculture. The everyday life of these fish workers is still dictated by fishing activity, even when it is blended within the larger social and residential context of a township that now has over 11 thousand inhabitants (INEGI, 2020).

These industrial and artisanal fishers, and the packers and loaders employed in El Sauzal harbor, maintain strong roots to this township and particularly to its labor dynamics and living spaces. Older fishers that retired from industrial fleets enter the harbor whenever possible to chat, while they smoke, with other fishers around the boatworkers. Fishers talk nostalgically of the days when families gathered to say goodbye whenever they left to sea, and waited to receive them when they returned. For these men it is inconceivable not to go to the harbor every morning, because it represents not only their lifelong workspace but their primary identity.

Less than 10 km away, the port of Ensenada is located in Todos Santos Bay (Fig. 1B) and began operations in 1877 following a marked increase in commercial trade with the United States. However, it was until the 1950s that Ensenada came to be considered an urban municipality (15,000 inhabitants) and reached the peak of its fishing activity (Padilla, 2016). The sardine fishery was developing in the Canadian and US Pacific coast from 1915 up to 1986, and the first cannery installed here (1933), La Industrial Ensenada, mainly packed sardine and mackerel in tomato sauce, with smaller quantities of abalone, tuna, squid, and mussel production (Ochoa, 1988). At least six additional canneries were installed between 1942 and 1958, with concurrent improvements in both fishing and packing gears and techniques, including the rapid expansion of the much higher-value tuna fishery.

Concurrently with the expansion of industrial fisheries and their social influence, the contiguous *Aduana* ('Customs,' after the customs office in the port) neighborhood was developed by artisanal fishers and their families starting in the 1930s. Neighborhood residents organized to build the first classrooms of their local primary school, which was also used to have neighborhood meetings to discuss issues such as water, sewage, and electricity services (Ochoa, 1988). According to Ochoa (1988), this neighborhood had informally but well-delimited spaces for fishers and their families, and families not engaged in fishwork; residents also occupied different spaces if they were artisanal or industrial fishers. The rhythms of everyday life were marked by fishing activities: if sardine was in season, men left for the port at night, artisanal fishers left early in the morning, and women waited for the "whistle" of the cannery when it was time to go work at the factory.

Nowadays, the *Aduana* is a marginalized neighborhood dealing with issues related to drug addiction and prostitution. It is still identified as a fishing neighborhood with resident artisanal fishers, and it is common to see fishing gear and *pangas* being repaired outdoors. Some have argued that issues of increased crime, drugs, and prostitution are a consequence of the embargo on tuna exports from Mexican fisheries imposed by the US in 1993; this bankrupted the entire industrial tuna fleet and caused a profound economic crisis from which the city continues to recover (Almaraz, 2018). Wealthier families in Ensenada were able to recover by investing in other industries such as local wine production (Almaraz, 2018) but job losses primarily affected fishers, their families, and those with livelihoods related to the harbor.

While women did not participate in industrial fishing in Ensenada or El Sauzal, they formed most of the labor force in the canneries and processing plants. This included women from the neighborhood, but also from surrounding areas with few alternative forms of employment, including Indigenous Kumiay youth from San Antonio Necua (40 km away from El Sauzal). Given that El Sauzal and the port of Ensenada are now within a large urban area, there has been an increase in other forms of employment for women aside from fishwork. An observation in this study is that current job advertisements from local processing plants seem to primarily request men, but the reasons for this must be further investigated.

The industrial and artisanal fishers that work here today live throughout the city of Ensenada and bike or drive to the harbor each morning. They continue taking some of their daily catch home and there is never a family celebration without a seafood dish. The current population of Ensenada is over 330,650 inhabitants (INEGI, 2020).

#### Bahía de Kino, Sonora

The locality of Bahía de Kino (Fig. 1 E) is divided in two towns: Kino Nuevo and Kino Viejo ('New,' and 'Old,' Kino). Kino Nuevo mainly includes vacation homes and some small residential neighborhoods, as well as hotels, shops, restaurants, a local museum, and the campus of University of Arizona's Prescott College. Kino Viejo (or simply Kino) originated as a fishing camp at the end of the XIX century, established by fishers from Puerto Peñasco (known as 'Rocky Point' in the US) and Indigenous cComca'ac (Comca'ac is the ethnonym of this Indigenous group but they are also known as the Seri, the name given by Spanish colonizers) to take advantage of the local abundance of totoaba (*Cynoscion macdonaldi*). In 1933, a local Lázaro Cárdenas Cooperative was created as an extension of its headquarters in Puerto Peñasco. In the 1940s, a local shark fishery developed due to the high demand for shark liver oil for nutritional purposes, but shark prices (and the fishery) declined once synthetic vitamins were developed, leaving the fishing site almost unpopulated.

The organization of fishing cooperatives and the proliferation of efficient outboard engines between 1965 and 1975 established Kino as a receptive place for national migrants from different states of Mexico, generally coming for work in the agricultural lands of nearby Hermosillo (the capital city of Sonora), but that ended up settling in the coastal town. While most began engaging in fishing activities, others started to diversify into handicrafts workshops, grocery stores and the sale of items for tourists.

During these years, groups of fishers from the southern state of Chiapas arrived while following shark migrations by land. These *chiapanecos* came to Kino and fished by associating with a local fishing permit holder who also helped procure room and board for them, commonly in a family house. Other groups that later started to arrive for extended stays in Kino were the *lobeños*, from Puerto Lobos

in the south of Sonora state. They also associated with local permit holders, but were not widely welcomed and entered into conflict with local fishers because, unlike the *chiapanecos* that fished sharks (a product that was no longer fished in Kino), the *lobeños* targeted the same species as *kineño* fishers. In the 1990s, conflicts further intensified between the coastal population and new immigrants, this time workers leaving the agricultural fields where they suffered terrible exploitation and labor abuses, and who contributed to increased fishing effort and a perceived decline in local resource availability in Kino (De la Torre and Sandoval, 2015).

The current population of Bahía de Kino is about 6500 inhabitants (INEGI, 2020) and the fishing activity continues to diversify, with almost 50 known target species (in addition to bycatch) including sierra (*Scomberomorus* spp.), jellyfish (*Stomolophus* spp.), swimming crabs (*Callinectes* spp.), shrimp (Penaeidae), stingrays (Batoideae), octopus (mainly *Octopus bimaculatus*), scallops (mainly *Atrina* spp.), finfish and mollusks (Fernández et al., 2018; De la Torre and Sandoval, 2015). There are 32 registered cooperatives and 41 permit holders, 365 *pangas*, 459 fishers and 506 permits (Fernández et al., 2018). Although the town extends inland, the houses located closer to the beach are mostly of fisher families.

In daily life, men are usually the ones that go fishing and diving in the morning, though there are a few but dedicated women who participate in capture fisheries. When *pangas* start to return, boys, girls, women, other men, and older people approach to help with the unloading of the catch, the gutting of fish, and the cleaning of the *panga*; fishers pay these informal workers back with fish. Women in particular also have a more active participation in fishing activities, managing the administration of some cooperatives and leading family cooperatives. Women commonly lead processing activities such as boiling crabs and removing and packing the flesh, filleting fish, and selling these products in the town or packaging them for transport to other markets. Women also primarily work in a small local processing plant cleaning, peeling, and packing shrimp, and many are employed in the tourism sector of the town as receptionists, administrators, and cleaning and sales. Kino Viejo can be defined as a fishing town, even though only about 20% of the population is occupied in fishwork

and many residents work in tourism-associated services (De la Torre, 2015).

#### Punta Chueca and El Desemboque de los Seri

The Indigenous Comca'ac people have habited the central coast of the state of Sonora since the prehispanic era. The knowledge and use of ocean and desert resources are not only the base of their subsistence, but the core of their culture and cosmogony. Like other Indigenous peoples in northern Mexico, their population was organized in nomadic groups and diminished drastically during the colonial period. With independence from the Spanish empire and a posterior national "whitening" policy, the Comca'ac continued being attacked by the Mexican state until their population was reduced to 130 persons at the beginning of the XX century (Luque and Doode, 2009).

Slowly, their population continues to recover. In the 1970s, an extension of *ejidal* and communal lands were "granted" to them by the federal government, limiting with this their territory and their access to fishing resources. The government goal of these property rights was "to increase the Seri's chances of survival by reducing the probabilities of future conflicts with other local fishers of different ethnic origin" (Basurto, 2006:189). The *ejido* El Desemboque and its annex Punta Chueca are the two main localities (Fig. 1D) in which the Comca'ac peoples live, including 1200 persons (Hernández, 2018). This *ejido* includes Tiburón Island and the Infiernillo Channel (Fig. 1A, D), and the Comca'ac have exclusive legal rights to resource use in the island and channel. Although the Comca'ac generate some income from the sale of artisanal handicrafts and basketry pieces (*coritas*), artisanal fisheries and the harvest of wild fruits and seeds from the desert are still central to their livelihoods, even with the intensive introduction of processed foods through their commercial networks.

Artisanal fisheries were mainly for internal trade and subsistence until the 1970s, when the commercial fishery of the *callo de hacha* (penshell; *Atrina* spp.) began. At this time, Comca'ac fishers did not have sufficient equipment to develop a commercial fishing activity, so outside fishers proposed to include all the Comca'ac men who wanted to fish in their crews and offered them *pangas*, fishing and diving gear,

and to buy their catch. The majority of the Comca'ac men agreed with these informal contracts, but some refused to allow outsiders to capture their resources (mainly penshell and swimming crab) inside their territory. This disagreement between Comca'ac people revived old conflicts associated with lineage and clan adscription and added to the difficulties in establishing their own Indigenous cooperatives. As the majority decided to let outside fishers into their maritime territory, a small but important part of the men remain excluded from the initial benefits of working with outside fishers.

In 1980, the parastatal fisheries support company PROPEMEX provided technical assistance and financing for *pangas*, outboard motors and fishing equipment, invested in seafood warehouses and employed Indigenous and outside fishers for scallop production. The company declared bankruptcy only four years later (Basurto, 2006), but with access to new equipment some of the Comca'ac men continued dive fisheries for different species up to the present, and a swimming crab trap fishery has been important since the 1990s (Delgado, 2009). Continuing Indigenous traditional practices, women and children have access to shoreline resources. During low tide, women and children use buckets and bags to collect shells and vertebrae for the elaboration of necklaces that they sell to visitors; they also look for octopuses and bivalves that are prepared that same day with fish in *sopa de mariscos* (seafood soup). The Comca'ac thus participate in daily subsistence and commercial fisheries that are still woven with cultural practices.

This area is ecologically productive due to persistent upwelling (Lavin and Marinone, 2003) and therefore known to regional fishers. Nowadays, fishers in these localities include Comca'ac peoples and non-Indigenous fishers with whom they have made agreements to fish in the Infiernillo Channel (just off Punta Chueca; Fig. 1). For instance, Basurto (2006) explains that relatively few Comca'ac men prefer to dive, so they allow outside divers to work with them. Comca'ac fishers still have a shortage of *pangas* and equipment, so outside fishers may be allowed within their territory for a fee. In other arrangements, buyers are obligated to pay the fuel for a Comca'ac fishing team to work at sea; highlighting complex tenure and power dynamics, crew may sometimes use the gas for other activities, or a fishing team leader may decide to sell catch to another buyer. Relationships between the

Comca'ac and outside fishers and traders are complex and have sometimes led to conflicts when agreements are breached by either side. Given the abundance of fish resources relative to the rest of the region, fishers from other localities such as Bahia de Kino Bay might enter (or try to enter) to fish illegally in Comca'ac territory; this has led to fights at sea and sometimes also the forcible seizure of catch (Delgado, 2009).

While Comca'ac women traditionally participated mainly in gleaning during low tides and in preparation of meals as outlined above, younger women have more recently begun engaging in commercial activities. This includes conservation work, ecotourism, and small-scale commercial fishing. Given the progressive empowerment of Indigenous youth throughout Mexico, it is likely that these young women will become new leaders in generational transitions in these Indigenous communities.

Even though the Comca'ac legally and traditionally control access to their territory and fishing resources, poverty, marginalization, and village isolation have contributed to an increase in the influence of outside fishers and buyers who eventually control and sell 90% of their production (Basurto, 2006). Comca'ac fishers can decide with whom they make an arrangement, but year-round fishing for penshell, driven by outside market demand, may lead to an overexploitation of this fishery and a terrible loss for Comca'ac economy and fishing resources in the long-term.

The Comca'ac people find themselves strongly linked to and partially dependent on outside forces, notably including Bahia de Kino and other outside fishers and buyers. For Comca'ac elders, the transformation of scallops, crabs, and other resources from their biocultural world and heritage into commodities, is only part of a domination by outsiders of Comca'ac territory, cosmogony, and culture. These fishers and buyers also control transportation, fishing equipment and *pangas*, market access, and prices for Comca'ac production. Ironically, although the scallop is one the most valuable seafood products in northwest Mexico, the conditions of the Comca'ac population in general are some of the most precarious in terms of infrastructure and basic well-being in Sonora state. The Comca'ac own fish resources within their territories, but only very few can buy their own equipment; this of course has led to ongoing conflicts between them and outsiders, and contributes to continuing self-segregation between these two groups.

## Conceptualizing types of fishing community

Based on the observations and information documented through the ethnographic research outlined above, we inform conceptualizations of different types of fishing communities among these case studies. We again stress that communities and relationships with fisheries are very diverse. An understanding of the types of communities derived from our field research can help us better inform fisheries research and policy, and these types may certainly be applicable to other localities. However, it is essential to consider the *questions* that can help researchers understand tangible and intangible relationships in their own locales.

Following from the analysis of the relationships and contexts of our case studies, we thus formulated three questions that together help understand and differentiate relationships with local capture fisheries and fishwork, both individually and communally. These are: Do fisheries and fishwork structure daily-life activities? Do fisheries and fishwork inspire and frame fundamental worldviews? Are fisheries and fishwork the primary source of employment and income? Re-analyzing each case study from the perspective of these questions then led to a framing of three types of fishing communities, as summarized in Table 1. In that context, the nuances and descriptions of each type are expended at length in the Discussion below.

## Discussion

Through these case studies we have shown the enormous heterogeneity that characterizes fisheries and their surrounding 'communities' (Jentoft et al., 2017), in a relatively limited geographical area in the northwest Mexico (Fig. 1). The ethnographies of these localities exhibit their heterogeneity and the risks of applying the term "fishing community" to loosely refer to a locality in which fishers live. Indeed, unspecified use detracts from characteristic fishing practices, the people who engage in them, and the fisheries management, research, and policy recommendations intended to support sustainable and equitable fisheries. Beyond efforts to incorporate local knowledge in fisheries research and policy (Beaudreau and Levin, 2014; Cisneros-Montemayor et al.,

**Table 1** Framing approach to understand and differentiate community relationships with capture fisheries and fishwork

| Fishing as... | Worldview                                                                                                                                                                                             | Daily life                                                                                               | Income                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Way of life   | Individual and group identity is fully enveloped within fishwork. In the case of Indigenous communities in particular, fisheries and seascapes are often integral to cosmogony (creation mythologies) | Daily routines and individual occupations and places within community revolve around fishwork            | Fishwork is often the primary (or only) source of income, but the practice of fishing and the relationships it underpins are seen as more important than the income itself |
| Livelihood    | Fishwork is often a part of more diverse ways of life, but the practice of fishing and experience of seascapes contribute to overall worldviews                                                       | Daily routines largely revolve around fishwork, but also incorporate other occupations and relationships | Fishwork is often the primary, but not only, source of income                                                                                                              |
| Job           | Fishwork provides unique experiences that may inform but do not underpin worldviews                                                                                                                   | Fishwork structures routines during working hours                                                        | Fishwork is often a seasonal or sporadic source of income (that can still be significant)                                                                                  |

Columns indicate bounding questions regarding the relative importance of fishwork in framing worldviews, structuring daily activities, and contributing to income. Rows include the types of communities identified for cases in this study, with table contents providing further context for each

2021), this more critical consideration of the nature of relationships with fisheries can better inform research and the policy recommendations that directly affect fishworkers and their practices.

Three important general insights emerged in our examination of cases under the scoping questions related to the importance of fisheries for worldviews, daily routines, and income (Table 1). First, some fishing communities can be continuous net recipients of new inhabitants, not socially closed-off localities. Second, even when fishing communities are relatively isolated geographically, most of them are widely and actively connected through the commercialization of their products in regional and international markets. Third, most fishing communities are not self-sufficient anymore; even those retaining relatively more traditional practices often sell at least some catch to buy other products. These and other factors do not preclude the fact that such communities remain firmly rooted in their seascapes (Brookfield et al. 2005), with face-to-face interactions and shared daily experiences, and these attributes should always be considered together with more quantitative indicators of the ‘importance’ of fisheries (Clay and Olson, 2008).

Although ‘fishing communities’ (e.g., Ejido Esteban Cantú) can organize separately and under different modalities, many exhibit a “cultural collective consciousness” based on shared daily experiences, that integrates an “us” and an “others” and that establishes ways of solidarity through face-to-face

relations. The fishing community has a set of values, a tradition, knowledge, referents, and pride that give the members cohesion and a desire to transmit this to their children. In such communities, we can find a sense of unity that transcends the personal and the fishing activity itself, and it is common to find kinship relations between their members. In practice, this may mean that programs aiming to support alternative livelihoods in such communities may not have the uptake that economics-based assumptions may anticipate. This has been documented in somewhat unsuccessful yet otherwise well-intentioned and well-funded programs throughout the world [e.g., Vietnam (Hanh, 2021), Kenya (Carter and Garaway, 2014)]. Alternatively, programs that are successful in economic and/or ecological terms may generate conflicts if they do not meet social goals related to local access to and participation in fishwork (e.g., in western Canada; Pinkerton, 2017). In an instrumental sense, any of these issues are crucial to avoid when resources are scarce and individuals are potentially already skeptical of new policy promises, though of course this is in addition to fundamental concerns on social equity and environmental justice.

Importantly, the boundaries that separate or unite individuals within communities are movable; on the contrary, they are so flexible that even fishing communities with a history of conflicts over resources can become a single ‘fishing community’ if they actively participate in cooperative management (Ayers and

Kittinger, 2014). Conversely, a ‘local community’ is formed by a geographically-bounded area or neighborhood (e.g., Ensenada-El Sauzal; Fig. 1B). Many interactions are still face-to-face, and though there is a more flexible delimitation between the *us* and the *others*, there are common interests and spaces that contribute to solidarity. A local community is nonetheless more heterogeneous than a ‘fishing community’ and tends to self-locate and identify as a geopolitical and administrative unit: a locality, a township, even an island. The core of an ‘Indigenous community’ is its ethnic adscription. In the case of the Comca’ac, for example, they are first and foremost an Indigenous community, that happens to be currently established in two localities. At each locality we can identify a local community and, within it, a fishing community. These two fishing communities are nevertheless linked, and it is possible that they could be considered one. Throughout our case studies, the *otherness* between fishing communities is not necessarily based on physical locality but is often based on competition for fishing resources (for example, the different welcomes experienced by *chiapanecos* in Kino Viejo compared to *lobeños*, based on their overlap with locals’ target species). Following from our key argument, proposed fisheries management or policy actions would clearly require a consideration of these various embedded groupings to ensure that they are appropriate in scale and target populations.

Drawing from the characteristics and dynamics of our case studies as outlined above, we derive and highlight three types of motivators among fishworkers. These types are certainly not intended to be universal, but rather illuminate that individuals within localities that host fisheries can have core differences in their practices and world-views that are essential to recognize as we envision and develop future fisheries research and policy (Cisneros-Montemayor and Cisneros Mata, 2018). The following discussion is therefore mostly focused on our case studies, though we do note parallels from related work in other localities. The first type, “fishing as a way of life,” represents fishing as a livelihood but also a collective cultural consciousness and a strong identity related with the sea and the practice of fishing. Here, fishing sustains social structures and is the most common source of occupation, and underpins food security and food sovereignty. The second, “fishing as livelihood,” reflects a strong attachment to fishing as an

occupation, but one that exists in the context of other activities such as agriculture or tourism that together contribute to work and income throughout the year. The third, “fishing as a job,” comprises those who work as fishers given need and opportunity but are readily willing to leave if presented with a better job opportunity. To be clear, we do not argue that one type is more desirable or laudable than another, but we do argue that researchers (and policymakers) must be aware of these or other motivations to ground their perspectives, and that this should be at the very least considered and briefly specified in fisheries literature aimed at informing policy.

### Fishing as a way of life

As we show, for a particular type of fishworker, fishing, gleaning, processing, and preparing seafood are not only a job or a form of subsistence. When fishing activities constitute ways of life, the people in it structure their world, their values, their daily family dynamics and routine, and their identity around the sea, coast and everything related to fishing. These relationships are often described for artisanal fishers (Jentoft et al., 2017), but this can also be the case for industrial fishers such as those in Ensenada’s harbors, whose lives depend completely on fishing incomes, their place on ships, and the schedule of at-sea work.

“We fishermen go to the sea, a month, three months or more...we leave everything here, the family, the wife, and we do not know what will happen because on the high seas we never know what might happen, but we leave because we cannot stay here, we do not know what will happen to us, but the sea calls us” (Industrial fisher, interview, 2015).

This is also the case of divers and fishers in Esteban Cantú that work all year by the sea, even if fishing products have low market prices and even when they have property rights on land or can work in more lucrative activities. One interviewed diver expressed that he cannot be away from the sea because he felt sick, he felt something was missing and no other thing he knew was as satisfactory as diving. These men and women develop, through experience, a reference system along the coast, an ecological knowledge and a very strong perception of clouds, tides, and even sounds:

“I don’t need to see the sea, in the morning I get up and just by listening to it I know if we are going to be able to go to work or if we are not going to be able to go [...] for so many years listening to the sea every day, it is as if the sea speaks to me, it speaks to us, what happens is that not everyone knows how to listen to it” (SPR Diver, Delgado, 2019:105)

In Bahía de Kino, this kind of fisher can operate within a cooperative organization, working with a permit holder, or can be a ‘free fisher’ that may be technically fishing illegally. Sometimes they fish just like their fathers and grandfathers who founded the first fishing camps in the 1930s. Sometimes, permit holders or cooperative members that do not fish anymore, because of their age or a health problem, nonetheless work in the organization of on-land fishing tasks to stay by the sea and within the fishing environment. In these cases, it is not the catch that determines identity, but a network of relationships grounded in fishing.

“I started fishing when I was 13 years old with my father who taught me, back in Puerto Lobos. Since then, I have been a fisher for almost 50 years. There was a time when I tried to be a farmer, because I was an *ejidatario*, but I never could leave the sea and now, old, ill, alone, there is no place I would rather be than here, fishing and walking by the beach” (Artisanal fisher, interview, 2000).

In the case of Comca’ac fishers in Punta Chueca, they have an additional historical and cultural background that defines the Indigenous group as one with a strong bond with the sea. Even their origin mythology refers to a turtle who created life and then created the Comca’ac people. Although they were displaced from their prehispanic territory and confined to a communal territory, their relationship with the sea and its resources is maintained. This can be seen in their commercial and subsistence fishing practices, in their cosmogony and in seasonal diets that primarily include fish, mussels, clams, sea snails, and octopuses (Delgado, 2009; Hernández and Narchi, 2018). A fisher-hunter-gatherer culture system characterizes Comca’ac people. Fishing not only defines their self-identity through the economic dimensions of fishing as an

industry—in which they fully participate—but also through biocultural and heritage dimensions.

These types of relationships are commonly associated with but not exclusive to Indigenous communities. For example, in southern England management and monitoring generally recognize industrial and small-scale (coastal) fisheries, yet local relationships between ‘proper’ and ‘part-time’ fishers are governed by complex socio-economic and cultural factors only superficially related to their choice of fishing gear (Ota et al. 2008). Here, ‘community identity’ might indeed seem to group based on trawling or netting practices, but the choice of gears is determined by different traditional ways of life and thus management focusing on technological characteristics alone does not address the ultimate factors driving fisher effort decisions.

### Fishing as a livelihood

Certainly not all fishers, divers and other persons involved in fishing activities have a historic and cultural background in fishing as a way of life, but this does not mean they do not have profound bonds with their work. Indeed, choices both to fully engage with or partially detach from fishwork may follow from much more recent occurrences. This could include, for example, the collapse of traditional local fisheries (e.g., in eastern Canada; Power et al., 2014), displacement from traditional areas and practices (e.g., through racist policies and/or coastal development; Mollet, 2014), or cultural practices that specifically alternate between fishing and other traditional livelihoods (e.g., in West Africa; Sene-Harper et al., 2019).

In Esteban Cantú, for instance, there are men and women who combine their work in fishing with other occupations related to their property rights in the *ejido* (e.g., administration, agriculture) or to other kinds of tasks related to ocean sectors. This way of life is diverse but shares common interests with the practice of fishing as well as and their physical environment.

“I started fishing as a child, because I saw my dad and he taught me to dive for lobster. The we set up the cooperative and formalized, right? And there I have always been able to dive, until today I have not had scares [...] When you are no longer young you realize

that you need to have another option, because you cannot dive all your life [...] that's why I started investing in the tourist field, because when I can no longer be by the sea, at least I will have another influx of money and something to do so as not to die of boredom at home" (Diver, Esteban Cantú, interview, 2013).

Others find other work during seasonal fishing closures, including as shop attendants, in mechanical workshops, as construction supervisors, or they migrate to work for some months in nearby California.

"During sea urchin season I work in fishing and then, when the season closes, I go to the other side [to the United States] to work with my brother in construction. I am a fisher but I go for a while to not get bored, to see my brother, and I take the opportunity to earn some dollars" (Artisanal fisher, Esteban Cantú, interview, 2013).

In the SPR, for example, these individuals are active members of the cooperative and have a voice and a vote in every decision of the organization as members of the *ejido*. If they leave during closed seasons, they may allow younger men to work their place in the SPR, sometimes asking for a fee payment in return, and sometimes without it.

In Bahía de Kino, despite being much more geographically isolated than the urban areas of El Sauzal or Esteban Cantú (Fig. 1), the fishing community is a lot larger and its composition more heterogeneous and complex. Fisheries dynamics involve formal and informal agreements, cooperatives and free fishers and divers, permit holders (employers), family cooperatives, and supporting businesses dependent on fishing activities. Fishing is at the core of livelihoods and truly shapes daily activities for many people in this town.

"I am, as they say, a free fisherman, I am not in a cooperative [...] I either work with the same permit holder, or I work with friends who let me get in the *panga* [...] I really like to go fishing, it is hard work but you will never be short of food" (free fisher, Bahía de Kino, interview, 2001).

Women in Bahía de Kino are also highly engaged in fishing and gleaning, and often work gathering clams and oysters along beaches and estuaries, which they sell to small restaurants or to individual buyers. Others work in seasonal processing plants cleaning and peeling shrimp or processing other seafood products. There is an increased participation of women in paid work in fishing and seafood processing all over northwest Mexico (Delgado, 2021), and they are continuously strengthening and shaping these fishing communities' livelihoods and social dynamics.

### Fishing as a job

For some fishworkers, being in this industry is just a job. They do not have specific cultural traditions in fishing or, if they do, they do not share the strong bond that other fishers and divers have with fishing and the sea beyond a source of income. Some men and women work as fishers and in processing when there is no other job or source of income, or during periods when processing plants make temporary hires to process large seasonal catches. In Esteban Cantú, for example, the SPR sometimes hires young men to work in someone's place as part of a crew and when the season closes, the hired fishers simply leave, or only work until they can find another job.

Bahía de Kino commonly receives day-laborers (both Indigenous and non-Indigenous migrants from other regions in Mexico) that abandoned their jobs in nearby agricultural fields and try working with a fishing crew. Sometimes these new fishers stay in fishing and get into the community "way of life," but other times, they quit because they do not like the sea, or get seasick, or scared if they do not know how to swim. Other men and women return each year to work in seafood processing plants to benefit from windfall seasonal incomes and leave for other seasonal or more permanent jobs elsewhere when seasons conclude.

These fishworkers very often develop deep knowledge and skills regarding fishing and processing, though not necessarily a particular attachment with fisheries or interest in the problems or challenges of the localities where they work, even when some of them return to fishing-related work for years.

## Concluding remarks

This study highlights the importance of providing content and context to the concept of ‘fishing community,’ and we propose two initial steps that would help better convey this meaning in research and, ultimately, policy. First, it is important to specify the type of grouping that underlies our bounding and use of community in research. Drawing from our case studies, this could include, for example, a local or island community (physical location), a community of fishworkers (shared practice), a joint-access community (shared legal status, e.g., an *ejido*), or an Indigenous community (Indigenous or ethnic group). Second, we must be clear about the motivation and depth of relationships of individuals in a community with the practice of fishing. Here this is typologized as *way of life*, *livelihood*, or *a job*, but researchers must consider these motivations (and boundings) in the context of their own research areas. As an entry point into these critical perspectives on a particular locality, guiding questions include whether fisheries and fishwork 1) structure daily-life activities; 2) inspire and frame fundamental worldviews; and 3) represent the primary source of employment and income.

Fisheries and fishworkers can be exceptionally heterogeneous and so cannot be expected to neatly fall within any typology. Nevertheless, we argue that considering—and explicitly describing—these general aspects of different ‘fishing communities’ is essential for understanding their socio-ecological systems and power dynamics, and for designing and implementing equitable solutions for management challenges.

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**Data Availability** The information generated and analyzed during the current study is available from the corresponding author on reasonable request.

## Declarations

**Conflict of interest** The authors have no conflicts of interest to report.

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